PoE to USB HID Bridge - Rev 4e (Übersicht)

Hierarchischer Schaltplan, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL, ERC 0/0).
Verbindungen zwischen den Blättern über globale Labels und Power-Symbole (GND, +3V3, +5V).
ISOLATIONSGRENZE 1500 V: PRIMÄR (PoE) = PSE_CT12/CT36/SP45/SP78, VDD_POE, VSS (J1-Kabelseite, D1-D3, C1, C14-Pin1, U1-Pins 4/5).
SEKUNDÄR = alle übrigen Netze. Überquerer: nur U1 (Ag9905-LPB), J1 intern (Übertrager/Bob-Smith), C14 (DNP).
DNP: C1, C14, R8, R9, R30, R31, C30, J3 (SWD-Randpads); U1 = Beistellteil (DNP im JLC-Export, Consigned).

poe_pd_module1

B2 poe_pd_module - PoE-PD-Modul Silvertel Ag9905-LPB

Mode-A/B-Brücken D1/D2 (MB6S) → TVS D3 SMAJ58A → U1 Ag9905-LPB (Cla...
Ausgang +5V_RAW: C11 100 µF Polymer + C12 22 µF + C10 → FB1 → +5V, ...
DNP-Optionen: C1 (Eingangs-C), R8 (Mindestlast RB), R9 (ADJ-Trimm),...

Datei: poe_pd_module.kicad_sch

ethernet_magjack1

B1 ethernet_magjack - RJ45-MagJack HR871150C

Kabelseite (PRIMÄR, PoE): J1.7-J1.10 → PSE_CT12/CT36/SP45/SP78 (zu ...
PHY-Seite (SEKUNDÄR): TX/RX-Paare mit 49,9-Ω-Terminierung, CT an +3...
MagJack-LEDs J1.11-14 unbenutzt (NC)

Datei: ethernet_magjack.kicad_sch

ldo_3v3_1

B4 ldo_3v3 - 3,3-V-LDO TLV75733P

+5V → +3V3 (1 A), C15 Eingang, C16 Ausgang

Datei: ldo_3v3.kicad_sch

mcu_w55rp20_1

B5 mcu_w55rp20 - W55RP20 (RP2040 + W5500)

Entkopplung C40-C60, FB2 → +3V3A, Quarze Y1 12 MHz / Y2 25 MHz, EXR...
SWD-Randpads J3 (1 SWCLK, 2 GND, 3 SWDIO, 4 +3V3) für Stiftleiste 2...

Datei: mcu_w55rp20.kicad_sch

usb_host_port1

B6 usb_host_port - USB-A-Host

TPS2553 Lastschalter (R25 44,2 k → 532-659 mA), FAULT offen; USBLC6...
R30/R31 (15 k) und C30 DNP

Datei: usb_host_port.kicad_sch

status_led1

B7 status_led - WS2812B-2020-V6 an +3V3

LED_DIN (GPIO28) → R38 330 Ω → D11 DI, C64 100 nF

Datei: status_led.kicad_sch

PWR_FLAG-Symbole (ERC) sitzen in den Unterblättern an je einem Pin der Netze: +3V3A, +5V, GND, VDD_POE, VSS

Isolationsgrenze: PRIMÄR = PSE_*, VDD_POE, VSS | SEKUNDÄR = Rest
Quelle: outputs/poe_usb_hid_bridge_rev4e.net (SKiDL), ERC 0/0
Rev 4e: Ag9905-LPB (SMT), WS2812B-2020-V6 an +3V3, SWD-Randpads J3, 6 Lagen 20 x 80 mm
poe_usb_hid_bridge

Sheet: /
File: poe_usb_hid_bridge_rev4e.kicad_sch

Title: PoE to USB HID Bridge

Size: A3	Date: 2026-10-23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 1/7

B2 poe_pd_module - PoE-PD-Modul Silvertel Ag9905-LPB

Mode-A/B-Brücken D1/D2 (MB6S) → TVS D3 SMAJ58A → U1 Ag9905-LPB (Class 0, 1500 V isoliert)
Ausgang +5V_RAW: C11 100 µF Polymer + C12 22 µF + C10 → FB1 → +5V, D12 OVP-Zener, TP9
DNP-Optionen: C1 (Eingangs-C), R8 (Mindestlast RB), R9 (ADJ-Trimm), C14 (Y-Kap VSS↔GND)
ISOLATIONSGRENZE: PRIMÄR = PSE_*, VDD_POE, VSS (U1 Pins 4/5) | SEKUNDÄR = +5V_RAW, AG_ADJ, GND, +5V (U1 Pins 1-3)



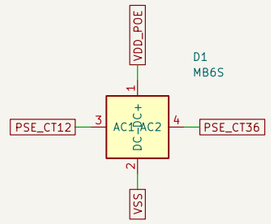


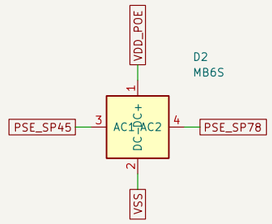


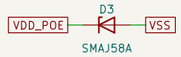












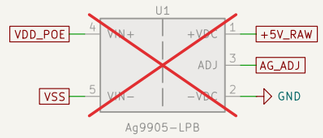












Netze über globale Labels / Power-Symbole verbunden (gleiche Namen = gleiches Netz)
Rev 4e, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL)
poe_usb_hid_bridge

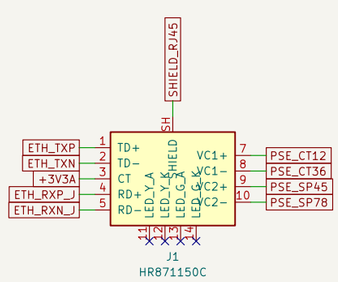
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File: poe_pd_module.kicad_sch

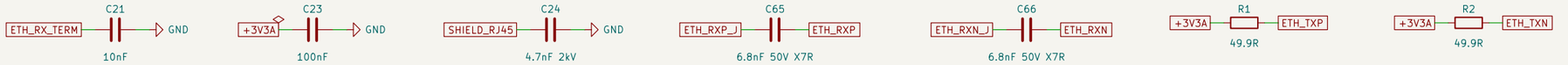
Title: PoE to USB HID Bridge - B2 poe_pd_module - PoE-PD-Modul Silvertel

Size: A3	Date: 2026-10-23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 2/7

B1 ethernet_magjack - RJ45-MagJack HR871150C

Kabelseite (PRIMÄR, PoE): J1.7-J1.10 → PSE_CT12/CT36/SP45/SP78 (zu B2)
PHY-Seite (SEKUNDÄR): TX/RX-Paare mit 49,9-Ω-Terminierung, CT an +3V3A, Schirm über R7 ‖ C24
RX-Koppel-C C65/C66 6,8 nF (DC-Trennung bei gemeinsamem CT, WIZnet-Referenz): J1-Seite ETH_RXP_J/ETH_RXN_J
MagJack-LEDs J1.11-14 unbenutzt (NC)







Netze über globale Labels / Power-Symbole verbunden (gleiche Namen = gleiches Netz)
Rev 4e, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL)
poe_usb_hid_bridge

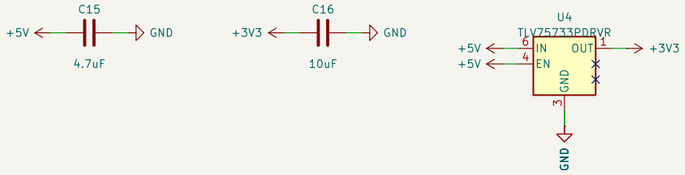
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File: ethernet_magjack.kicad_sch

Title: PoE to USB HID Bridge - B1 ethernet_magjack - RJ45-MagJack HR8711

Size: A3	Date: 2026-10-23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 3/7

B4 ldo_3v3 - 3,3–V–LDO TLV75733P

+5V → +3V3 (1 A), C15 Eingang, C16 Ausgang



Netze über globale Labels / Power–Symbole verbunden (gleiche Namen = gleiches Netz)
Rev 4e, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL)

poe_usb_hid_bridge

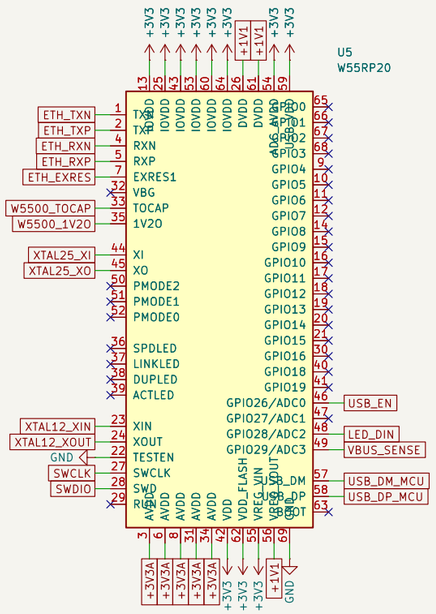
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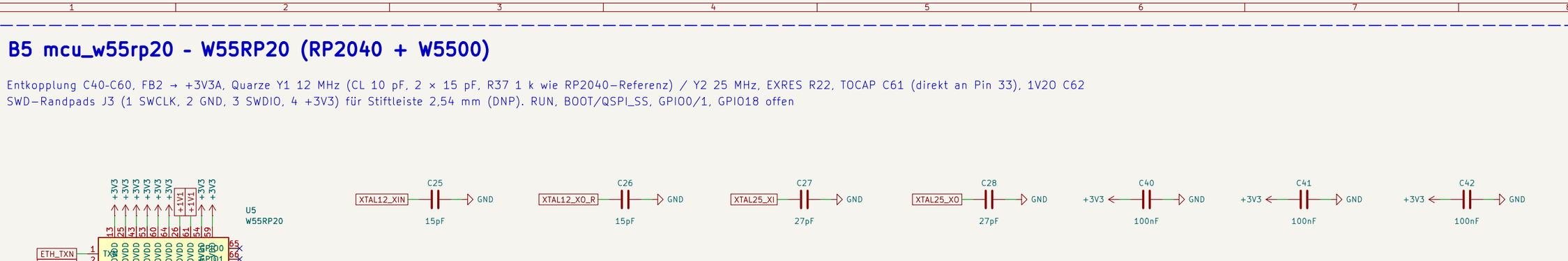
Title: PoE to USB HID Bridge - B4 ldo_3v3 - 3,3–V–LDO TLV75733P

Size: A3	Date: 2026–10–23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 4/7

B5 mcu_w55rp20 - W55RP20 (RP2040 + W5500)

Entkopplung C40-C60, FB2 → +3V3A, Quarze Y1 12 MHz (CL 10 pF, 2 × 15 pF, R37 1 k wie RP2040-Referenz) / Y2 25 MHz, EXRES R22, TOCAP C61 (direkt an Pin 33), 1V2O C62
SWD-Randpads J3 (1 SWCLK, 2 GND, 3 SWDIO, 4 +3V3) für Stiftleiste 2,54 mm (DNP). RUN, BOOT/QSPI_SS, GPIO0/1, GPIO18 offen





Netze über globale Labels / Power-Symbole verbunden (gleiche Namen = gleiches Netz)
Rev 4e, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL)

poe_usb_hid_bridge

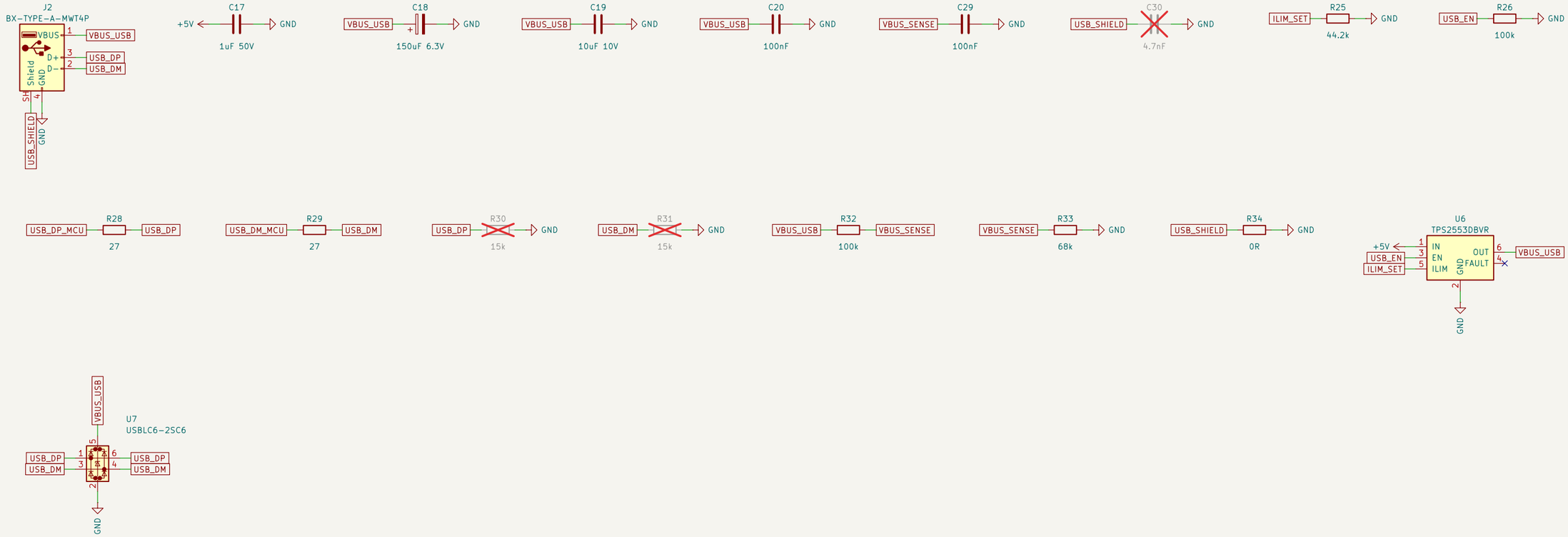
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File: mcu_w55rp20.kicad_sch

Title: PoE to USB HID Bridge - B5 mcu_w55rp20 - W55RP20 (RP2040 + W55

Size: A3	Date: 2026-10-23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 5/7

B6 usb_host_port - USB-A-Host

TPS2553 Lastschalter (R25 44,2 k → 532-659 mA), FAULT offen; USBLC6 ESD, 27-Ω-Serien-R, VBUS-Teiler
R30/R31 (15 k) und C30 DNP



Netze über globale Labels / Power-Symbole verbunden (gleiche Namen = gleiches Netz)
Rev 4e, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL)
poe_usb_hid_bridge

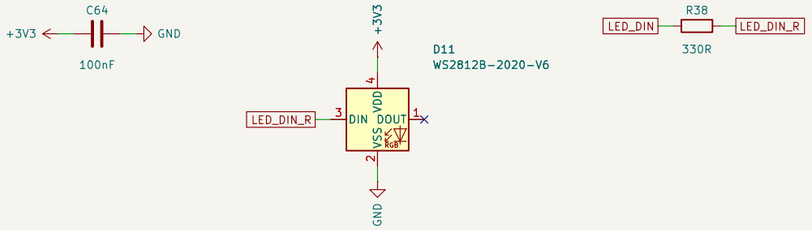
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Title: PoE to USB HID Bridge - B6 usb_host_port - USB-A-Host

Size: A3	Date: 2026-10-23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 6/7

B7 status_led - WS2812B-2020-V6 an +3V3

LED_DIN (GPIO28) → R38 330 Ω → D11 DI, C64 100 nF



Netze über globale Labels / Power-Symbole verbunden (gleiche Namen = gleiches Netz)
Rev 4e, generiert aus outputs/poe_usb_hid_bridge_rev4e.net (SKiDL)
poe_usb_hid_bridge

Sheet: /status_led1/
File: status_led.kicad_sch

Title: PoE to USB HID Bridge - B7 status_led - WS2812B-2020-V6 an +3V3

Size: A3	Date: 2026-10-23	**Rev: 4e**
KiCad E.D.A. 10.0.6		Id: 7/7